

Akshat Ramachandran

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Research Overview

Vision

Design the next generation of AI inference infrastructure by treating efficiency as a first-class design objective. My work focuses on cross-layer co-design across model compression algorithms, runtime systems, and specialized hardware architectures to ensure the future of AI scales through efficiency, not just compute.

Interests

Computer Architecture, Efficient AI, Hardware-Software Co-Design, Foundation Models (LLMs and VLMs), Computer Vision, Computer Arithmetic, Digital Circuit Design

Languages and Tools

Python, C/C++, Verilog, SystemVerilog, HLS (Vivado, Catapult), CUDA, Triton, ML frameworks (PyTorch, Keras), HW design tools (Synopsys Design Compiler, Innovus)

Education

Georgia Institute of Technology Aug 2023 – May 2027
Ph.D. in Electrical and Computer Engineering | Atlanta, GA GPA: 4.0/4.0
Minor: Finance (Scheller College of Business)
Thesis: Co-Designing Hardware and Algorithms for Efficient Foundation Model Computing via Adaptable Sparsity and Quantization
Advisor: Dr. Tushar Krishna

Georgia Institute of Technology Aug 2023 – Dec 2025
M.S. (Thesis) in Electrical and Computer Engineering | Atlanta, GA GPA: 4.0/4.0
Thesis: Algorithm-Hardware Co-Design of Digital Compute-in-Memory Architecture Supporting Flexible and Temporal N:M Sparsity
Advisor: Dr. Tushar Krishna

Veermata Jijabai Technological Institute (VJTI) Aug 2018 – May 2022
B.Tech in Electronics Engineering | Mumbai, India GPA: 9.80/10.0
Department Rank: 1/83 (Gold Medallist) | Institute Rank: 2/480
Thesis: Next Generation Architecture for Computer Vision
Advisor: Dr. John Gustafson (Arizona State University)

Professional Experience

NVIDIA Research May 2026 – Present
Ph.D. Research Intern, Accelerators and VLSI (Manager: Dr. Brucek Khailany) | Santa Clara, CA

- Developing an algorithm-hardware co-design framework for automatic mixed-precision quantization.

NVIDIA Research May 2025 – Aug 2025
Ph.D. Research Intern, Accelerators and VLSI (Manager: Dr. Brucek Khailany) | Santa Clara, CA

- Proposed ThinKV, a thought-adaptive KV-cache compression technique that exploits the heterogeneous importance of reasoning tokens to enable selective quantization and token eviction for efficient long-context reasoning.
- Achieved near-lossless accuracy while reducing KV-cache memory usage by more than 96%.
- Filed a patent and led work resulting in an ICLR 2026 oral presentation.

Intel Labs

Aug 2024 – Dec 2024

AI Hardware Research Intern (Managers: Dr. Arnab Raha, Dr. Souvik Kundu) | Santa Clara, CA

- Proposed a layer-wise outlier-density-aware N:M sparsity selection method and FlexCiM, a digital compute-in-memory architecture with flexible structured sparsity support.
- Integrated FlexCiM into Intel NPU-v7, resulting in an ISLPED 2025 Best Paper Award and a patent filing.

NVIDIA Corporation

May 2024 – Aug 2024

Architecture Energy Modeling Intern (Manager: Dr. Visu Subramanian) | Santa Clara, CA

- Developed an AI-based power estimation tool for NVIDIA GPUs.
- Leveraged transfer learning to distill knowledge from parent chips to derivative architectures, reducing data collection and training effort by 5 $\times$.

Georgia Institute of Technology

Aug 2023 – Present

Graduate Research Assistant (Advisor: Dr. Tushar Krishna) | Atlanta, GA

- Conducting doctoral research on co-designing hardware and algorithms for efficient foundation model computing via adaptable sparsity and quantization.

Lemurian Labs

Jun 2023 – Aug 2023

Hardware Intern (Managers: Dr. Ashish Kaul, Dr. Theodore Omtzigt) | Menlo Park, CA

- Designed and verified a smart NoC to interlink functional units in the Lemurian tensor engine using SystemVerilog and Python/PyUVM, achieving a 35% increase in operating frequency and 2.5 $\times$ better performance over the existing interconnect.
- Spearheaded deployment of an adaptive logarithmic datatype in Python and C++, and evaluated mixed-precision quantization of NumPy-implemented BERT and GPT, yielding a 10 $\times$ speedup over floats.

Samsung Research & Development

Aug 2022 – Jul 2023

Visual Intelligence Engineer (Managers: Dr. B.V. Vandrotti, Mr. Jooyoung Kim) | Suwon, South Korea

- Contributed to accelerator and algorithm co-design, optimization, and research-to-commercialization of depth perception and processing solutions for XR devices, leading to an ICIP publication, a patent, and a Samsung Best Paper Award nomination.
- Co-designed and cross-validated a deep-learning-based ToF depth fusion algorithm and SoC architecture in Python and HLS, enabling a 67% reduction in on-device inference latency.
- Designed a systolic dataflow accelerator in SystemVerilog for ToF acquisition, reducing latency by 60%.

Indian Institute of Technology Bombay

Jul 2021 – Dec 2022

Research Assistant (Advisor: Dr. Madhav Desai) | Mumbai, India

- Designed and verified a reconfigurable, programmable, multi-thread 40 Gb/s NIC and MAC architecture in HLS and Verilog, interfaced as a memory-mapped peripheral to the $2 \times 4 \times 32$ Ajit processor in a MeiTY-funded project.
- Implemented custom packet generation and forwarding flows for verification and NIC support libraries in C.

Samsung Research & Development

May 2021 – Jul 2021

Summer Research Intern (Manager: Dr. Ankit Dhiman) | Bangalore, India and Suwon, South Korea

- Researched memory-efficient deep-learning algorithms and architectures for AR applications.
- Remodelled Samsung's 3D reconstruction pipeline in Python and C, and designed a memory controller that extended k-means-style clustering to parameters, achieving nearly 2 $\times$ higher performance than prior pipelines.

Publications (* Equal Contribution)

Conference Publications

ThinKV: Thought-Adaptive KV Cache Compression for Efficient Reasoning Models

Akshat Ramachandran, Marina Neseem, Charbel Sakr, Rangharajan Venkatesan, Bruce Khailany, and Tushar Krishna

International Conference on Learning Representations (ICLR), Apr 2026

Oral presentation (Top 1.13% of accepted papers)

OuroMamba: A Data-Free Quantization Framework for Vision Mamba

Akshat Ramachandran*, Mingyu Lee*, Huan Xu, Souvik Kundu, and Tushar Krishna

International Conference on Computer Vision (ICCV), Oct 2025

Accelerating LLM Inference with Flexible N:M Sparsity via a Fully Digital Compute-in-Memory Accelerator

Akshat Ramachandran, Souvik Kundu, Arnab Raha, Shamik Kundu, Deepak A. Mathaikutty, and Tushar Krishna

International Symposium on Low Power Electronic Design (ISLPED), Aug 2025

Best Paper Award

MicroScopiQ: Accelerating Foundational Models through Outlier-Aware Microscaling Quantization

Akshat Ramachandran, Souvik Kundu, and Tushar Krishna

International Symposium on Computer Architecture (ISCA), Jun 2025

Acceptance rate: 23%

AIRCHITECT v2: Learning the Hardware Accelerator Design Space through Unified Representations

Jamin Seo*, Akshat Ramachandran*, Yu-Chuan Chuang, Anirudh Itagi, and Tushar Krishna

Design, Automation and Test in Europe Conference (DATE), Mar 2025

CLAMP-ViT: Contrastive Data-Free Learning for Adaptive Post-Training Quantization of ViTs

Akshat Ramachandran, Souvik Kundu, and Tushar Krishna

European Conference on Computer Vision (ECCV), Oct 2024

Algorithm-Hardware Co-Design of Distribution-Aware Logarithmic-Posit Encodings for Efficient DNN Inference

Akshat Ramachandran, Zishen Wan, Geonhwa Jeong, John Gustafson, and Tushar Krishna

Design Automation Conference (DAC), Jun 2024

Acceptance rate: 19%

PositIV: A Configurable Posit Processor Architecture for Image and Video Processing

Akshat Ramachandran, John Gustafson, Anusua Roy, Rizwan Ahmed Ansari, and Rohin Daruwala

Euromicro Conference on Digital System Design (DSD), Aug 2023

Journal Publications

GPU Based Building Footprint Identification Utilising Self-Attention Multiresolution Analysis

Rizwan Ahmed Ansari*, Akshat Ramachandran*, and Winnie Thomas

All Earth, 2023

Workshop Publications

Polestar-Cache: Reconciling Parallel Decoding and Accuracy in Diffusion LLMs via Token Drift-Aware KV Cache Recalibration

Mingyu Lee*, Akshat Ramachandran*, Souvik Kundu, and Tushar Krishna

Workshop on Latent & Implicit Thinking – Going Beyond CoT Reasoning @ ICLR 2026

Recap: Training-Free Compensation for Coarse Activation Channel Pruning in Compressed LLMs

Mingyu Lee*, Akshat Ramachandran*, and Tushar Krishna
Machine Learning for Computer Architecture and Systems @ ISCA 2025

Preprints

OneDSE: A Unified Microprocessor Metric Prediction and Design Space Exploration Framework

Ritik Raj, Akshat Ramachandran, Jeff Nye, Shashank Nemawarkar, and Tushar Krishna
ArXiv

Patents

Think Before you Quantize: Thought-Aligned Adaptive Quantization via Algorithm-Hardware Co-Design for Efficient Large Reasoning Models

Akshat Ramachandran, Marina Neseem, Charbel Sakr, Rangharajan Venkatesan, and Brucek Khailany
Provisional patent filed based on work at NVIDIA Research

Compute-in-Memory Architecture for Accelerating Neural Network Operation

Akshat Ramachandran, Arnab Raha, Souvik Kundu, Shamik Kundu, and Deepak Mathaikutty
US patent filed based on work at Intel Corporation

A Method and System for Acquiring, Processing and Monocular Depth Completion/Estimation of Sparse Time-of-Flight Sensors

Akshat Ramachandran, Ankit Dhiman, Basavaraja Shanthappa Vandrotti, Jooyoung Kim, and Lokesh Rayasandra Boregowda
US Patent, Oct 2024 (US20240331176A1), based on work done at Samsung Research & Development

Awards and Honors

MLCommons Rising Star, Cohort featuring emerging leaders of ML and systems research	2026
Best Paper Award, ACM/IEEE International Symposium on Low Power Electronic Design (ISLPED)	2025
2 nd Place, Student Research Competition, IEEE/ACM International Symposium on Microarchitecture (MICRO)	2024
DAC Young Fellow, ACM/IEEE Design Automation Conference (DAC)	2024
Samsung Best Paper Award Nominee, Top 5% of advanced research at Samsung	2023
Samsung Spot Award, For contributions to AR/VR algorithms on Samsung Galaxy devices	2022
Best Undergraduate Thesis Award	2022
Professional Level SW Competency, Global Samsung Aptitude Test	2022
2 nd Place, Electrothon Hackathon	2022
2 nd Place, Vision Beyond Limits Competition	2021
2 nd Place, E-Yantra International Robotics Competition	2021

Talks

“Thought-Adaptive KV Cache Compression for Efficient Reasoning Models”

NVIDIA, Santa Clara, CA (Host: Marina Neseem) Dec 2025

“Outlier-Aware Quantization: ViTs, LLMs and SSMs”

Tambe Lab, Stanford, CA (Host: Dr. Thierry Tambe) Mar 2026

A*STAR, Singapore (Host: Dr. Himeshi De Silva) Jan 2026

Apple, Cupertino, CA (Host: Dr. Fernando Mujica)	Aug 2025
“Designing Quantized Accelerators for AI”	
Google TPU Team, Sunnyvale, CA (Host: Dr. Krishnakumar Nair)	Jul 2025
NVIDIA, Santa Clara, CA (Host: Dr. Marina Neseem)	Jul 2025
ACM Student Research Competition @ MICRO	Nov 2024
“Learning the Hardware Accelerator Design Space through Unified Representations”	
EuroSys-ASPLOS 2025	Mar 2025
“Logarithmic-Posit Encodings”	
IBM Research, New York, NY	Jun 2024
Lemurian Labs, Menlo Park, CA (Host: Dr. Theodore Omtzigt)	May 2024
“Deep Learning Model Quantization”	
NVIDIA, Santa Clara, CA (Host: Dr. Visu Subramanian)	May 2024
CoCoSyS Liaison Meeting, Online (Host: Emily Watson)	Mar 2024

Media Coverage

Neptune.AI: Impressive research on efficient foundation model inference	2025
OpenVINO: Q2’25: Technology Update – Notable Research – ThinKV	2025
OpenVINO: Q4’25: Technology Update – Notable Research – OuroMamba	2025
Georgia Tech: Ramachandran won second place at the ACM Student Research Competition	2025
CoCoSyS: Director’s Note:CoCoSyS Top-5 Accomplishments in Year 2, Quarter 3	2025
Georgia Tech: Research conference spotlight, International Conference on Computer Vision	2025
ACM SIGARCH: MICRO 2024 Trip Report: Success at Scale	2024

Grants

Research Program (Student Contributor)

DARPA Semiconductor Research Corporation (SRC) JUMP 2.0 CoCoSyS Center (2025–2026)

Student Travel Grants

ICLR 2026, ISCA 2025, MICRO 2025, and DAC 2024; total support of approximately \$6,000

Academic Service

Submission Co-Chair

MICRO 2026

Technical Program Committee Member (Conferences and Journals)

CoNGA 2024, ECCV 2024, SCOPE-ICLR 2025, IEEE COINS 2025, IEEE TCAD, ICCV 2025, NeurIPS 2025, CoNGA 2025, ICLR 2026, MLSys 2026, IEEE CAL, IEEE TIP, ICML 2026, BMVC 2026, NeurIPS 2026, IEEE TCSVT

Artifact Evaluation Committee

HPCA 2024, ASPLOS 2025, HPCA 2026, ISCA 2026

Teaching Experience

Hardware for Machine Learning (GT ECE 8803)

Spring 2026

Guest Lecturer on Efficient AI, Prof. Divya Mahajan

Hardware for Machine Learning (GT ECE 8803)

Spring 2025

Guest Lecturer on AI for Hardware, Prof. Tushar Krishna

Select Student Advisees

Daniel Moon Undergraduate intern from SNU, South Korea	2026–Present
Mingyu Lee GT undergraduate visiting student from KAIST, South Korea Next destination: GT PhD Fall 2026	2024–2026
Pradyut Ganesh Undergraduate student from BITS Pilani, India Next destination: CMU MS Spring 2026	2025

References

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